

BARRE KNAPP, VT, USA

WMO: 726145

Lat: 44.204N

Lon: 72.562W

Elev: 343

StdP: 97.27

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94705

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.5	(c) -20.7	(d) -27.8	(e) 0.3	(f) -22.8	(g) -25.1	(h) 0.4	(i) -19.7	(j) 10.4	(k) -6.4	(l) 9.2	(m) -6.9	(n) 2.2	(o) 320	(p) 0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	29.4	21.0	27.9	20.1	26.5	19.4	22.5	27.2	21.5	26.0	20.6	24.7	3.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	16.3	24.8	20.0	15.4	23.9	19.1	14.5	23.0	67.6	27.3	64.1	25.8	61.0	24.8	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.7	7.8	7.0	-27.9	32.0	3.1	1.5	-30.1	33.0	-31.9	33.9	-33.6	34.7	-35.9	35.8
			-28.1	24.4	3.0	1.5	-30.3	25.5	-32.0	26.4	-33.7	27.3	-36.0	28.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-8.2	-6.7	-1.9	5.4	12.2	17.0	19.6	18.6	14.7	8.1	1.7	-4.4
	DBStd	10.92	7.23	6.22	5.91	4.94	4.27	3.70	2.85	3.00	4.05	4.52	5.29	5.91
	HDD10.0	2380	563	468	370	156	24	1	0	0	7	93	252	448
	HDD18.3	4497	821	701	626	390	194	67	19	34	122	318	498	706
	CDD10.0	1074	1	0	2	17	94	212	297	267	147	33	4	1
	CDD18.3	149	0	0	0	1	6	28	58	43	13	1	0	0
	CDH23.3	1331	0	0	2	17	79	291	489	336	112	4	0	0
	CDH26.7	273	0	0	0	3	13	71	108	59	21	0	0	0
(14) Wind	WSAvg	2.6	2.8	2.9	3.2	3.2	2.8	2.4	2.1	1.9	2.1	2.5	2.8	2.6
(15) Precipitation	PrecAvg	875	54	50	58	62	82	87	78	96	80	74	79	71
	PrecMax	1203	175	159	94	124	195	188	150	272	180	140	144	160
	PrecMin	625	5	10	10	19	10	35	27	34	17	5	34	28
	PrecStd	149	37	28	22	27	44	39	28	52	38	35	30	37
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	11.5	19.0	25.7	28.5	31.2	31.4	30.5	29.4	23.8	18.3	12.2
		MCWB	8.9	7.4	12.4	15.4	18.8	21.5	22.9	21.9	20.9	16.7	13.8	9.0
	2%	DB	7.6	7.7	12.6	21.1	25.7	28.7	29.3	28.2	26.3	20.7	15.1	8.4
		MCWB	5.6	4.3	7.5	12.2	16.4	20.2	21.4	20.6	19.7	15.1	11.7	6.4
	5%	DB	4.0	4.5	9.1	17.5	23.0	26.7	27.7	26.8	24.2	18.1	12.6	5.8
		MCWB	2.2	2.0	5.0	10.3	15.2	19.1	20.4	19.9	18.4	13.8	9.7	4.0
	10%	DB	1.6	2.2	6.4	14.0	20.8	24.6	26.2	25.2	22.0	15.7	9.8	3.3
		MCWB	0.3	0.3	3.1	8.1	14.1	18.0	19.7	19.3	17.3	12.2	7.0	1.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.8	8.7	12.5	16.5	20.0	23.1	24.1	23.3	22.4	18.7	15.0	9.9
		MCDB	11.2	10.2	17.7	23.0	26.2	28.1	29.1	28.0	27.1	21.2	17.3	11.9
	2%	WB	5.3	4.7	8.5	13.6	18.4	21.7	22.7	21.9	20.7	16.3	12.4	6.6
		MCDB	7.0	7.1	11.8	19.4	23.2	27.0	27.4	26.1	24.8	19.0	14.8	8.2
	5%	WB	2.3	2.3	5.5	11.3	16.9	20.4	21.7	21.0	19.5	14.4	10.0	4.1
		MCDB	3.7	4.2	8.5	15.9	21.3	24.8	26.0	25.0	23.0	17.7	12.4	5.5
	10%	WB	0.4	0.5	3.5	8.9	15.2	19.1	20.7	20.1	18.1	12.5	7.2	1.9
		MCDB	1.6	2.0	6.1	13.8	19.5	23.0	24.6	23.9	21.1	15.0	9.4	3.2
(35) Mean Daily Temperature Range	MDBR		9.2	10.0	9.7	10.8	11.6	11.3	11.2	11.3	11.3	9.9	8.4	8.0
	5% DB	MCDBR	10.7	11.6	13.4	17.3	15.6	14.5	13.1	12.9	13.7	13.9	12.5	9.8
		MCWBR	9.2	9.1	9.1	9.9	8.1	7.2	6.3	6.5	7.6	8.8	9.4	8.5
	5% WB	MCDBR	9.7	11.0	11.9	14.7	12.6	12.1	11.2	11.0	11.4	11.9	11.3	9.5
		MCWBR	8.7	9.1	8.8	9.4	7.3	6.6	6.0	6.0	6.8	8.4	9.4	8.5
(40) Clear-Sky Solar Irradiance	taub		0.271	0.280	0.316	0.346	0.368	0.419	0.430	0.403	0.357	0.326	0.302	0.272
	taud		2.382	2.331	2.361	2.401	2.394	2.248	2.225	2.291	2.413	2.511	2.513	2.453
	Ebn at Noon		861	915	922	918	903	853	838	849	866	855	822	827
	Edh at Noon		75	95	106	112	116	135	137	123	101	80	67	64
(44) All-Sky Solar Radiation	RadAvg		1.45	2.31	3.44	4.29	4.96	5.39	5.53	4.99	4.02	2.38	1.52	1.08
	RadStd		0.17	0.24	0.23	0.53	0.56	0.49	0.41	0.26	0.30	0.26	0.19	0.10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+71	N/A

Nomenclature: See separate page